GY462—Real Estate Finance

Lecture 9

1

Organizational remarks

- Soon upload on public folder 'mock examination'
  - Based on one past exams
  - Can solve it and hand in solutions
- How to get feedback?
  - Book my office hours: I will correct solutions and discuss them with you
  - Not officially graded, not compulsory, not part of final grade, but useful!
  - 'Highlights' discussed during ST revision session

2

Organizational remark – *GY462 Final Exam*

- Exam consists of 2 questions (both have equal weight)
- Answer both questions
  - One more 'theory', one more calculations
- Financial calculators are allowed
  - Know how to use it! Not xl…

3

Preview of Contents — *Lecture 9*

Debt Securitization

- The Evolution of the Secondary Market
- Residential Mortgage-Backed Securities (RMBS) Market
- Relevant Risks
- Types of RMBS & Allocation of Risks
- Concluding thoughts on RMBS-Market

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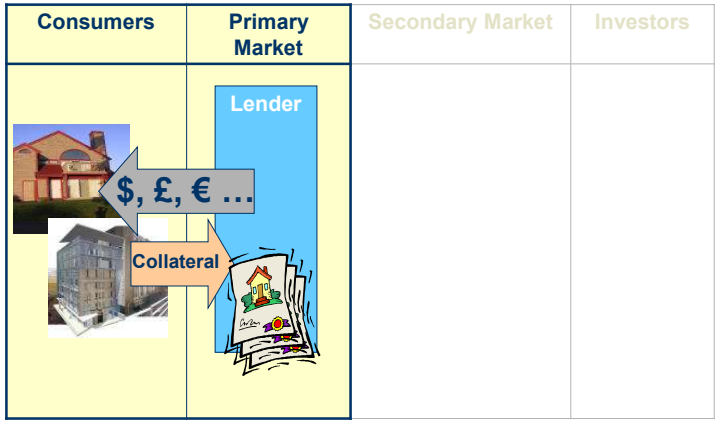
Starting Point: The Simple Life of a Mortgage



Why Has Secondary Mortgage Market Emerged?

- **Great Depression in US (1929~1939)**
  - **One third out of work**
  - **One third of banks closed**
  - **US 'country of renters' (~45%)**
  - **Standard loan terms in US**
    - **At least 40-50% money down**
    - **Interest only – no build up of equity**
    - **3-5 year term, balloon payment, refinance risk**

Why Has Secondary Mortgage Market Emerged? *Continued*

⇨ **Crisis in the mortgage market**

- **50% foreclosure rate in US during Great Depression**
- **Inability or unwillingness of private lenders to provide a reliable supply of mortgage credit**

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Why is Secondary Mortgage Market a 'Solution'?

1. Provides mechanism to replenish funds
   - Specialized mortgage originators can sell mortgages and thereby replenish funds
   - Facilitates geographic flow of funds
2. Allows shifting of risks from lenders to investors
3. Allows returns to specialisation

When Secondary Mortgage Market Emerged?

- Crisis & needs of lenders and investors lead to 'national commitment to housing'
  - 1938 Creation of FNMA (later: Fannie Mae)
  - 1968 FNMA split into Fannie Mae and Ginnie Mae; Fannie Mae privatised
  - 1970 Creation of Freddie Mac to create competition
  - 2008 Fannie and Freddie nationalised (bail-out)!
  - 2013 Fannie and Freddie 'doing better'
  - 2026 …?
- What exactly do 'Government Sponsored Enterprises' (GSEs) do?

What did Fannie and Freddie originally do?

- Fannie original role was
  - Enhance secondary MKT operations in federally insured or guaranteed loans
  - Manage loans and liquidate properties if needed; special-assistance programs
- Later became private enterprise
  - Still with a back-stop option for US treasury to acquire up to 2.25billion of notes
  - Implicit Gov't guarantee, allows Fannie to borrow at lower rates

What did Fannie and Freddie originally do? *Continued*

- And Freddie?
  - Federal Home Loan Mortgage Corporation
- Initial role of Freddie
  - Provide a secondary MKT (and hence liquidity) for conventional MGGs
  - Plus provide competition-type incentives for Fannie

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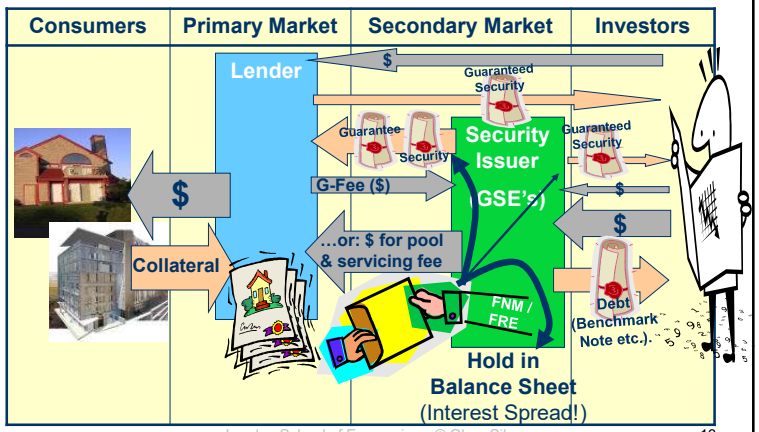
The 'Normal' Life of a Typical Mortgage in the US Today



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How does it work? *'Cash Programme'*

- **Focus on functioning of GSE**
  - **Other issuers of securities: big commercial banks**
  - **Use similar mechanisms**
- ***Alternative 1 – 'Cash programme':***
  - **GSE buys pool for $ and pays servicing fee then…**
  - **… decides whether to pool mortgages and use them as collateral for new GSE MBS or hold the mortgage loans as an investment**
    - **If it keeps mortgages: makes money from spread (interest on mortgage & interest on benchmark notes)**
    - **If it sells MBS to investors, has to guarantee the timely payment of principal and interest on the securities**

How does it work? *'Swap Programme'*

- ***Alternative 2 – 'Swap programme':*** **GSE does not keep repackaged securities to sell but…**
- **… gives securities back to lender (*'swap'*) for guarantee-fee:**
  - **GSE guarantees only (assumes credit risk) → MBS**
  - **GSE guarantees & repackages & creates tranches → CMO**
- **Vast majority of MBS created by GSE come through their swap programmes**

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How Do Security Issuers Fund Mortgage Purchases?

- **Major debt instruments of GSEs**
  - Long term, medium term and short-term notes
  - **Benchmark Notes (or Reference Notes)**
    - **Used instead of issuing medium term notes**
    - **Mimics characteristics of US Treasury Notes**
    - **Issued in large amounts to increase liquidity**
    - **Typically interest 25bp above Treasury Notes**
  - Foreign currency bonds

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What Risks Need to be Managed by Security Issuer?

1. **Interest rate risk**
2. **Credit risk (or default risk)**
3. **Prepayment risk**
4. Political risk
5. Liquidity risk

Interest Rate Risk

- **Does interest received on mortgages exceed interest paid on debt?**
- **Manage by keeping duration gap within narrow band**

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Duration

- **Measures the <u>weighted average time until cash flow repayment</u>**
- **Formula:**

$$Duration = \left[\frac{1 \text{ x } PV(CF_1)}{V}\right] + \left[\frac{2 \text{ x } PV(CF_2)}{V}\right] + \ldots + \left[\frac{N \text{ x } PV(CF_N)}{V}\right]$$

V...Total value of bond (or mortgage)
$PV(CF_t)$... Present value of cash flow in time period t

Duration Gap

- **Difference (in months) between duration of assets and liabilities**
- **Measures how well 'matched' are time flows of cash from assets and time flows of cash from liabilities**
- **If interest rates change unexpectedly: danger of widening 'duration gap'**
  - ***Why? Discuss…***

How Do Security Issuers Manage the Duration Gap?

- **Buy more or less mortgages**
- **Call back or issue more debt with appropriate terms**
- **'Callable Debt'**
  - **Call option allows to pay debt off early**
  - **Helps manage prepayment risk!**

Credit Risk (or Default Risk)

- **Evaluate on three factors**
  - **Capacity**
  - **Credit history**
  - **Collateral**
- **Credit enhancements (insurances)**

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Credit Risk — *Continued*

- **Complex default risk models**
  - **Based on capacity, credit history and collateral assessment**
  - **FICO score measures reputation**
  - **Automated**
- **House price decline and risk models**
  - **Look at HP decline risk in different locations/for different types of property**

Prepayment Risk

- **Prepayment allows homeowners to minimize interest rate risk**
- **Creates risk for mortgage owners that borrowers prepay mortgage during periods of falling interest rates**
  - **Cash out refinancing boom in US**

⇨ **Makes predicting CF from mortgage pool & pricing of mortgage pool difficult**

Other Risks – *Discuss*

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⇨ **Main Types of Securities?**

1. **Mortgage Backed Bonds**
2. **Mortgage-Pass-Through Securities (MPTs)**
3. **Collateralized Mortgage Obligations (CMOs)**

(Others: MPTB and Derivatives)

Mortgage-Backed Bonds (MBS)

- **Usually, include mainly residential mortgages**
  - Issuer establishes pool of MGGs + issue bonds to investors
  - Issuer retains ownership of MGGs
- **Usually, over-collateralized to ensure payments**
  - Between 125 and 240% over-collateralized
  - Meaning? MGGs in pool have outstanding balances larger than dollar amount securities are issued for
  - Issuers bear risks of default and prepayments
- **Usually, bonds have fixed coupons and specific maturities**

Pricing of an MBS – *An example*

- **Consider coupon *i=8%*; *$200m* MBB against *$300M* in pool**
- **Consider bond for *$10,000*, *10-years* maturity**
- **Annual interest: *$ 800*; principal at maturity: *$10,000* (whole amount)**
- **If require rate of return (RR) is *9%*, bond has PV of *$9,358*…**
  - **Bond is selling at discount: *93.58%* of par value**
  - **If RR *7%*, sells at premium; if RR *8%*, sells at par**

Mortgage Pass-Through Securities (MPTs)

- **A MPT is a security backed by a pool of many individual mortgages**
  - **Usually, pool grouped by type of MGG: homogeneous interest/maturity/security**
  - **Backed by the CFs and collateral from the underlying mortgages**
  - **'MBS' if issued by FNM; 'Participation Certificate' if issued by FRE**
  - **Other originators can 'pool' too: work with approved security underwriters and dealers**

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MPTs — Why 'Pass-Through'?

- **Servicer collects loan payments from homeowner and 'passes through' payments to investors each month**
  - **Investor receives monthly CF minus servicing and guarantee fees**
  - **Pass through of prepayment and interest rate risk to investor as well!**
- **Note: different from MBB, paying interests; this is passing through PMT/CF**

Pricing of MPT – *An example*

- **Consider MPT of *$1 million* with *10%* fixed interest rate**
- **Pass-trough carries a coupon of *9.5%***
  - ***0.5%* difference between *9.5%* and *10%* is servicing fee (on outstanding balance)**
- **Maturity is 10 years; 40 securities**

Pricing of MPT: CF table

Year	Pool Balance	P+I PMTs	Guarant.+ Service Fees	Total PMT to single Investors
0	1,000,000			(25,000)
1	937,255	162,745	5,000	3,944
2	868,235	162,745	4,686	3,951
3	792,313	162,745	4,341	3,960
4	708,799	162,745	3,962	3,970
5	616,933	162,745	3,544	3,980
6	515,881	162,745	3,085	3,992
7	404,724	162,745	2,579	4,004
8	282,451	162,745	2,024	4,018
9	147,950	162,745	1,412	4,033
10	0	162,745	740	4,050

Pricing of MPT: some mechanics – *Continued*

- **If require rate of return (RR) in MKT is identical to coupon (*9.5%*)…**
- **… price received by issuer = overall loan balance (selling at par)**
  - **CF discounted at 9.5% is $25,000 per security**
- **If RR=*8.5%*? Security will sell at a premium → *$26,130***
  - **If RR=*10.50%*? Sell at discount → *$23,944***

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Pricing of MPT and prepayments

- Offering price and coupon are usually specified when MPT issued
- Reason for specifying price is that investors might want RR different from coupon (based on MKT conditions)
- Other important reason: prepayment
  - Issuers take this into account → affect offering price

Pricing of MPT and prepayments – *Continued*

- How to forecast prepayments? Some methods:
  - Average maturity
  - Constant rates
  - PSA model

Collateralized Mortgage Obligations (CMOs)

- Mortgage-backed security where payments on underlying collateral are partitioned to create different maturity classes (tranches)
- Investors buy one or more tranche, with each tranche presenting a different maturity
- Investors receive payments of interest or principal prioritized according to tranche

CMOs and MPTs

- CMOs and MPTs are 'pay through' securities in that all amortization and prepayments flow through to investors
  - Security holder assumes prepayment risk in both cases
  - CMO is a multiple security class, mortgage pay-through security

⇨ But CMO modifies <u>how</u> risk is allocated…

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CMOs vs. MPTs

CMOs	MPTs
▪ Issuer retains ownership of mortgage pool and issues bonds as debt against mortgage pool ▪ Issuer has equity position	▪ Investors own individual interest in entire pool ▪ Issuer has no equity position
▪ Issued in multiple maturity classes (tranches)	▪ No tranches
▪ Prioritisation with respect to interest, principal and prepayment proceeds	▪ All investors assume prepayment and reinvestment risk

CMO mechanics illustrated

- Consider CMO: *$72mio* issued against *$75mio* of pooled MGGs
  - Difference is over-collateral
- MGGs: *10-year* maturity + *11%* RR
- 3 different 'tranches' (classes) made from this investment
  - A bond, coupon *9.25%*, *$27mio*, maturity *2-5yrs*
  - B bond, coupon *10%*, *$15mio*, mat. *4-7yrs*
  - Z bond, coupon *11%*, *$30mio*, mat. *6-10yrs*

CMO mechanics – *Continued*

- Weighted RR paid to bonds: *10.14%*
  - Different from *11%*: source of profit earned by issuer
  - Can make profit from packaging and 'targeting' risk preferences (+ fees)
  - Over-collateral: profit is made on equity invested in venture; 'insure' PMTs
- 3 different 'tranches' are paid with different priorities

➢ *How does it work?*

CMO mechanics – *Continued*

- Sequential payout…
  - *Tranche A:* gets own interest + interests from Z + *priority* in receiving principal + prepayments
  - *Tranche B:* till A is paid, receives interest only, then all principals and prepayment
  - *Tranche Z:* nothing till A + B paid, interest accruing on investment (compounded at coupon rate), then gets all interests + principals…

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CMO Mechanics in Tables – *Overall Structure*

Year	Pool Balance	P+I PMTs	Amort.	Interests	Owed to Holders
			-3.000=		
0	75.000				72.000
1	70.514	12.735	4.485	8.250	67.514
2	65.536	12.735	4.978	7.756	62.536
3	60.010	12.735	5.526	7.209	57.010
4	53.876	12.735	6.133	6.601	50.876
5	47.067	12.735	6.808	5.926	44.067
6	39.509	12.735	7.557	5.177	36.509
7	31.120	12.735	8.389	4.346	28.120
8	21.809	12.735	9.311	3.423	18.809
9	11.473	12.735	10.336	2.399	8.473
10	0	12.735	11.473	1.262	0

CMO Mechanics in Tables – *Tranche A payments*

Year	Owed to Holders	All Principal + Int. from Z	Coupon Int.	Total Payments
		= Amortization + 11% × Owed amount Z		
0	27.000			
1	19.214	7.785	2.497	10.282
2	10.573	8.641	1.777	10.418
3	0.981	9.592	0.978	10.570
4	0	0.981	0.091	1.072
5	0	0	0	0
6	0	0	0	0
7	0	0	0	0
8	0	0	0	0
9	0	0	0	0
10	0	0	0	0

Residual to close position

= Owed amount × 9.25%

CMO Mechanics in Tables – *Tranche B payments*

Year	Owed to Holders	All Principal + Int. from Z	Coupon Int.	Total Payments
		= Amortization + 11% × Owed amount Z – Amount to close A		
0	15.000			
1	15.000	0	1.500	1.500
2	15.000	0	1.500	1.500
3	15.000	0	1.500	1.500
4	5.334	9.665	1.500	11.165
5	0	5.334	0.533	5.867
6	0	0	0	0
7	0	0	0	0
8		0		0
9		0		0
10	0	0	0	0

Residual to close position B

= Owed amount × 10%

CMO Mechanics in Tables – *Tranche C payments*

Year	Owed to Holders	Interest on Z	Accrued Int.	Principal Allocation	Total Payments
		= 11% × Owed amount	Not paid, to A/B…	= Amortization period 5 – Amount to close B	
0	30.000				
1	33.300	3.300	3.300	0	0
2	36.963	3.663	3.663	0	0
3	41.028	4.065	4.065	0	0
4	45.542	4.513	4.513	0	0
5	44.067	5.009	0	1.474	6.484
6	36.509	4.847		7.557	12.405
7	28.120	4.016		8.389	12.405
8	18.809	3.093		9.311	12.405
9	8.473	2.069		10.336	12.405
10	0	0.932		8.473	9.405

Regular Amort.

Residual, close Z

Less than amort., rest to issuer!

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CMO Mechanics in Tables – *Residual Equity CF*

Year	Total Cash Flow	Total Payments A+B+Z	Residual CF to Equity
0			(3.000)
1	12.735	11.782	0.952
2	12.735	11.918	0.816
3	12.735	12.070	0.662
4	12.735	12.237	0.497
5	12.735	12.351	0.383
6	12.735	12.735	0.330
7	12.735	12.735	0.330
8	12.735	12.735	0.330
9	12.735	12.735	0.330
10	12.735	9.405	3.300
IRR=	*20.19%*		

CMO Mechanics – *Some Remarks*

- Can add prepayments and redo calculations
  - Please, look at B&F book! Important to understand flow of payments
- Important: cash flow to different tranches
  - A only at the beginning
  - B low at beginning, picks up around year 3
  - Z only from year 5
  - Residual equity: low, smooth PMT throughout, peak at end

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⇨ **Who should benefit from MBS and why?**

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Benefits of Mortgage-Backed Securities

- **Benefits to lender**
  - Asset on balance sheet is now liquid
  - Replenishment of funds for new loans
  - Spreading risk from lender to investor side
  - Outsourcing of credit risk management (specialisation)

Benefits of Mortgage-Backed Securities — *Continued*

- **Benefits to investors**
  - Secondary market provides mortgage related products that appeal to risk profiles of certain types of investors
  - Indirect investment in real estate
- **Benefits to homeowner**
  - Lower interest rates (up to ~50bp)
  - Savings of up to $10,000 over life of typical loan (slightly old estimates)

Concerns? *Discuss*

- **Implicit government backing for Fannie Mae & Freddie Mac**
  - Too big to fail? Apparently so…

⇨ **Market distortions, systemic risk, hidden information**
  - Subprime mortgage and the Great Financial Crisis (GFC)

Concluding remarks

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Conclusions

1. Investments in direct RE increasingly relevant for investors
2. Real estate becoming more operational and focussed on development
3. But RE markets are 'local markets' and require local market knowledge

Conclusions – *Continued*

4. Great opportunities for sophisticated investors in domestic and international RE
5. More difficult times ahead for traditional investors
   - More likely to experience disappointments
6. Conclusion: good job you took this course (even better with GY457, GY458 & GY472 ☺)!

Subprime Mortgages and the GFC

Self-study materials

[For interested students only; not part of the exam materials]

Subprime Mortgages: What and Who?

- What are subprime mortgages? Loans to people with low probability of repayment
  - Bad credit rating score
  - High DTI (55%+) and LTV ratios (85%+)
  - Poor documentation (Alt-A)
- Who were subprime borrowers? Important question when assessing risk of underlying loans

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Subprime Borrowers: What Characteristics?

- Higher interest rates and higher credit risk
  - Subprime borrowers have low FICO and high LTV compared to 'prime'
- Demographics? More likely to be Black and Hispanic, have low median income, and poor education (few college graduates)
  - Less financially informed/literate
- Also living in more 'risky' neighbourhoods
  - Lower average income

Subprime Borrowers Characteristics – *Continued*

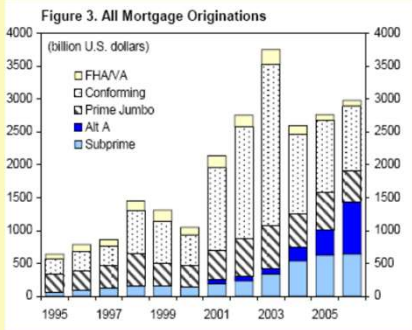
- Why did they turn to subprime MGGs?
  - Faced hurdles in getting normal MGGs: e.g. being approved or providing documents
  - Turned to less favourable loans as they become uncertain about chances of qualifying
  - Look primarily for loan approval, compared to prime borrowers
  - Not concerned about interest rate charges
- All in all, subprime categories is 'special': more at risk in case of adverse economic conditions

Subprime Mortgages: Why and How?

- Why did subprime MKTs developed?
  - 'Originate and distribute' models at the heart of these developments
- Development of underwriting process and securitization reduced costs of entering subprime mkts
  - Statistical models for credit risk should have provided reliable information
  - Securitization dispersed the risk of subprime

Subprime Mortgages: More Details

- By 2006, >50% of loans did not meet 'standard' criteria
- Subprime share of loans increased substantially
- Why? Expanding house prices stretched affordability
- But originators thought secondary MKTs could still help hedging risk



Source: IMF, 2007

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What Are the Characteristics of Subprime Mortgages?

- **Crucial feature: subprime MGGs closely linked to house price dynamics**
  - Give a loan to low-income families for a short period at affordable rate, while equity builds up
  - Then mortgage can be refinanced with lower loan-to-value ratio
- **After initial 2-3yr period, step-up rate**
  - Borrowers *have* to refinance because of spike in rate is very high

Characteristics of Subprime Mortgages – *Continued*

- **Who decides about refinancing? Lender!**
  - Based on whether house price have increased
  - Lenders are 'long' in real-estate, and safe only if prices keep going up
- **Example: ARM sub-prime MGG – "2/28"**
  - Originate in 2006, with initial rate at 8%; at reset date (after 2 years): LIBOR + spread → 12%
  - Teaser first, floating rate afterwards…
  - Reset is high: incentives for borrower to refinance
  - Note: around 80% of subprime have high prepayment penalties…

Characteristics of Subprime Mortgages – *Continued*

- **Why are house prices so important?**
  - If house prices increase → LTV of new loan down: less risky (relation between LTV and default)
  - But if prices go up too much, why don't borrowers walk away and extract equity? Penalties
  - Why not longer (normal) MGGs?
  - Risky borrowers: lender wants option to terminate/foreclose as early as possible (if needed)
  - Note: most borrowers refinance with same lender because of ties and fees…

Subprime Mortgages and House Prices – *Continued*

- **If house prices increase, convenient for lender to refinance**
  - Less risky, larger collateral, larger asset if future default
- **If house prices flat or decreasing?**
  - Lender in a position to decide what to do: decision to default is transferred from borrower to lender
  - If lender does not want to refinance most likely option is borrower will default
  - Can't pay reset interest, difficult to borrow elsewhere: if prices going down → borrower even more risky

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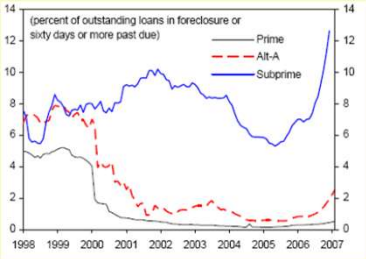
Subprime Mortgages: What Happened?

- Expanding trend continued till early 2007
  - More appetite for risk (mis-priced?) by investors
  - Anecdotal evidence of borrowers lying about financial resources to access increasingly less affordable housing
- In 2007: higher interest rates (inflation risk) and flattening or decreasing house prices
  - Difficult for subprime borrowers to repay loans or refinance: higher (reset) rates
  - Default → start selling homes, further deflating prices
  - Higher unemployment rates in some areas exacerbated situation

Subprime Mortgages: What Happened? – *Continued*

- Delinquency rates jumped
- Highest default rates for 'affordability products'
  - ARMs with option, IO only, negative amortization, teaser rates…
- 90% of subprime included (risky) affordability options…



Source: IMF, 2007

Subprime Securities: Some Specific Features…

- Subprime securities (SS) particularly tricky. Why?
  - Cash flow largely comes from payments of MGGs through ***refinancing***
  - However, cash flows depend on 'triggers' linked to prepayments and defaults of MGGs
  - Triggers can divert cash flows from one tranche to another tranche
  - Sometimes can erode protection for higher tranches – which were priced as 'safe'
- More details…

Subprime Securities (SS) Specific Features

- SS have over-collateralization (OC), but…
  - OC is created with <u>excess spread</u> (XS), <u>not usually part of the initial deal</u>
  - XS = remaining interest after paying regular interests to bondholders
  - Furthermore: there are OC targets; only once OC targets are met, XS can be used to pay residual holders
- Other features?
  - Step-down date, performance triggers and Available Fund Caps (AFC)

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SS Specific Features – *Continued*

- Step-down date
  - Prior to this date, senior bonds receive 100% of principal pre-payments
- Triggers
  - Usually delinquency and loss triggers; if triggers 'fail', payments to mezzanine & subordinate bonds are stopped
- Available Funds Cap
  - Bonds usually pay fixed coupon, whereas underlying MGG are usually ARMs
  - To avoid paying out more than cashing in, there are often caps

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SS Specific Features: Why It Matters

- Typical SS deal: tranches size as % of collateral are very small
  - Yet highly rated (BBB or above) by agencies
  - Why? Expectation is that underlying MGGs refinance properly, so credit enhancements build up…
  - After step-down date and using XS, subordinate trances will look fine
- This all depends on refinancing, i.e. on house prices going up!

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Subprime Securities: Is This All?

- Next link with Collateralised Debt Obligations (CDOs)
  - Special vehicles: buy portfolios of fixed income assets
  - Issue different tranches of risk in market
  - CDOs have underlying portfolios including RMBs or CMOs
- Other feature: CDOs often managed
  - Manager allowed to trade limited amount because assets amortize…
  - To achieve maturity need to trade, e.g. take some cash paid into CDO and reinvest it

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More on CDOs

- Why were CDOs attractive even if they included subprime?
  - SS tranches often thin and difficult to sell: further package them into CDOs
  - SS-related 'info' would disappear into CDOs
  - No data about exact CDO exposure to subprime: need to look at all underlying securities…
  - Yet exposure ex-post was large: 50% of high grade CDO bonds and 77% of mezzanine…

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Where Did These CDOs End-Up?

- Difficult to say but significant category is SIVs/conduits
  - Off-balance sheets vehicles purchasing medium/long term fixed income assets issuing short term paper
  - Highly leveraged
  - Conduits tend to be covered by committed liquidity lines with financing banks
  - SIVs use models approved by agencies to manage liquidity, but no explicit liquidity lines

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Complexity, Loss of Information and Crisis…

- In a nutshell, a 'Chinese boxes' game
- Subprime risk everywhere, but no one knows with certainty where…
  - Can't answer simple question such as how exposed is my CDO or SIV?
  - Huge loss of information/asymmetric info
- When house prices flattened/refinancing stopped, defaults began/crisis rolled out

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